Industry 4.0 and Beyond - A Summary

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Outline

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- WiFi 7 (802.11be)
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- Summary

Overview

Our paper covered the foundations of industry 4.0 and examined the future technologies that will play a part in establishing the next generation of industrial networking (industry 5.0).

Through analyzing the technologies presented in this paper (5G, WiFi 7, and TSN applications) we can provide a foundational understanding of CPN concepts and relate these concepts to future innovations in the world of industrial technology.

Industry 4.0 and Beyond

Industry 4.0 saw use of robotics and networked devices to facilitate the process of industrial automation. As industrial technology expands and is redefined in industry 5.0, strong enhancements to industry 4.0 technologies need to be made this include the growth of 5G and WiFi 7.

Goal for Industry 5.0: Enhance productivity, flexibility, and human-machine collaboration as robotics, artificial intelligence, and vehicular automation technology expands.

Industry 4.0 and Beyond

Key Points from Our Paper:

- Further integration of Cyber Physical Systems (CPS), Industrial Internet of Things (IIoT), and Artificial Intelligence (AI) into one ecosystem is needed
- 5G, WiFi 7, and TSN will form the communication backbone of Industry 4.0/5.0
  - Reliable, low-latency communication is vital for evolution of automation and robotics

Time Sensitive Networks

Overview:

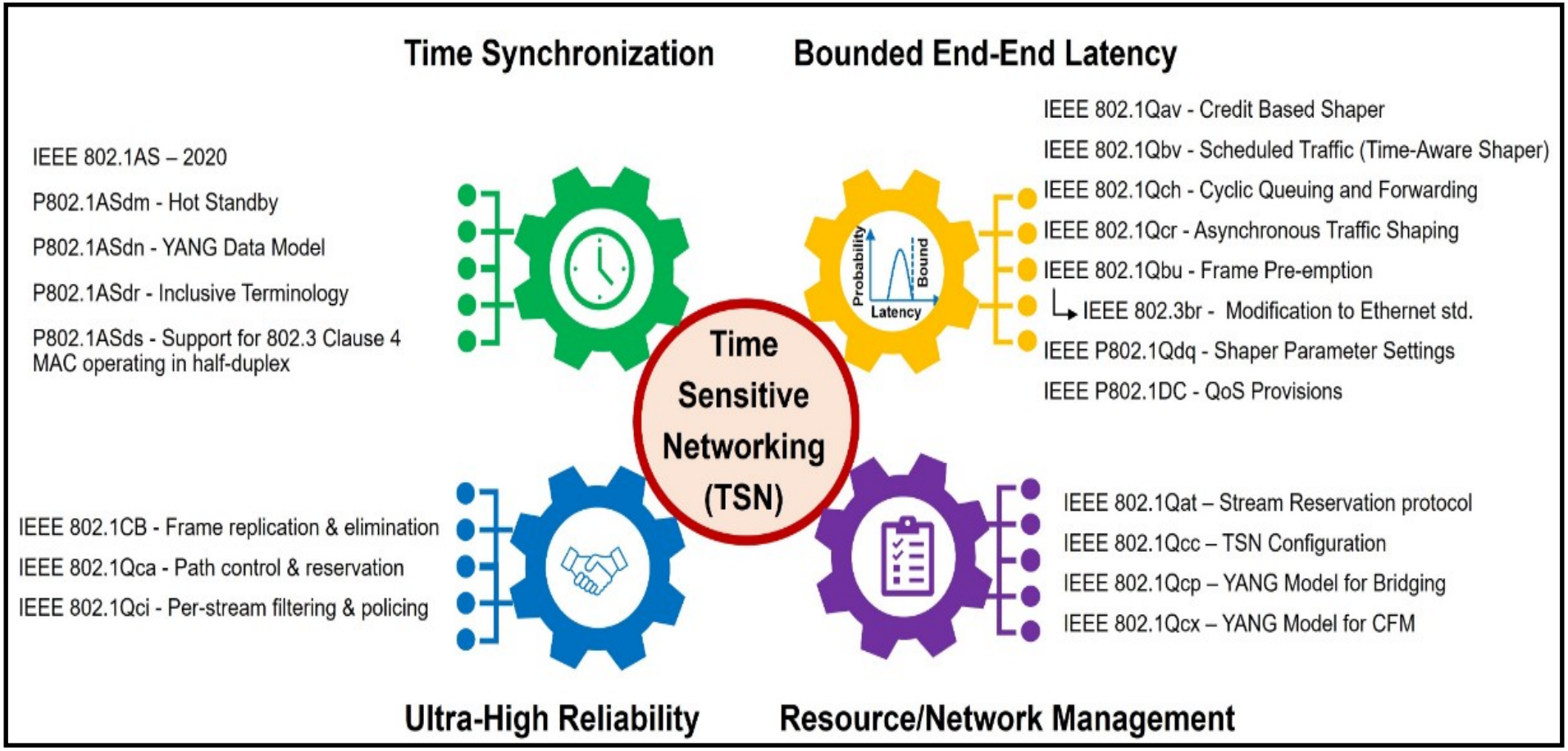
Networked environments where extremely low/zero packet loss is necessary and bounded, end-to-end latency, is required.

- All networked elements MUST be globally synced for deterministic behavior
- **OPC-UA** is the defacto standard protocol at the **control level**
- **PROFINET** widely used at the **field level**

Applicable Standards:

- IEEE 802.1AS – Time Synchronization
- IEEE 802.1 Qbv – Bounded Latency
- IEEE 802.1CB – Reliability

Time Sensitive Networks - Standards Expanded



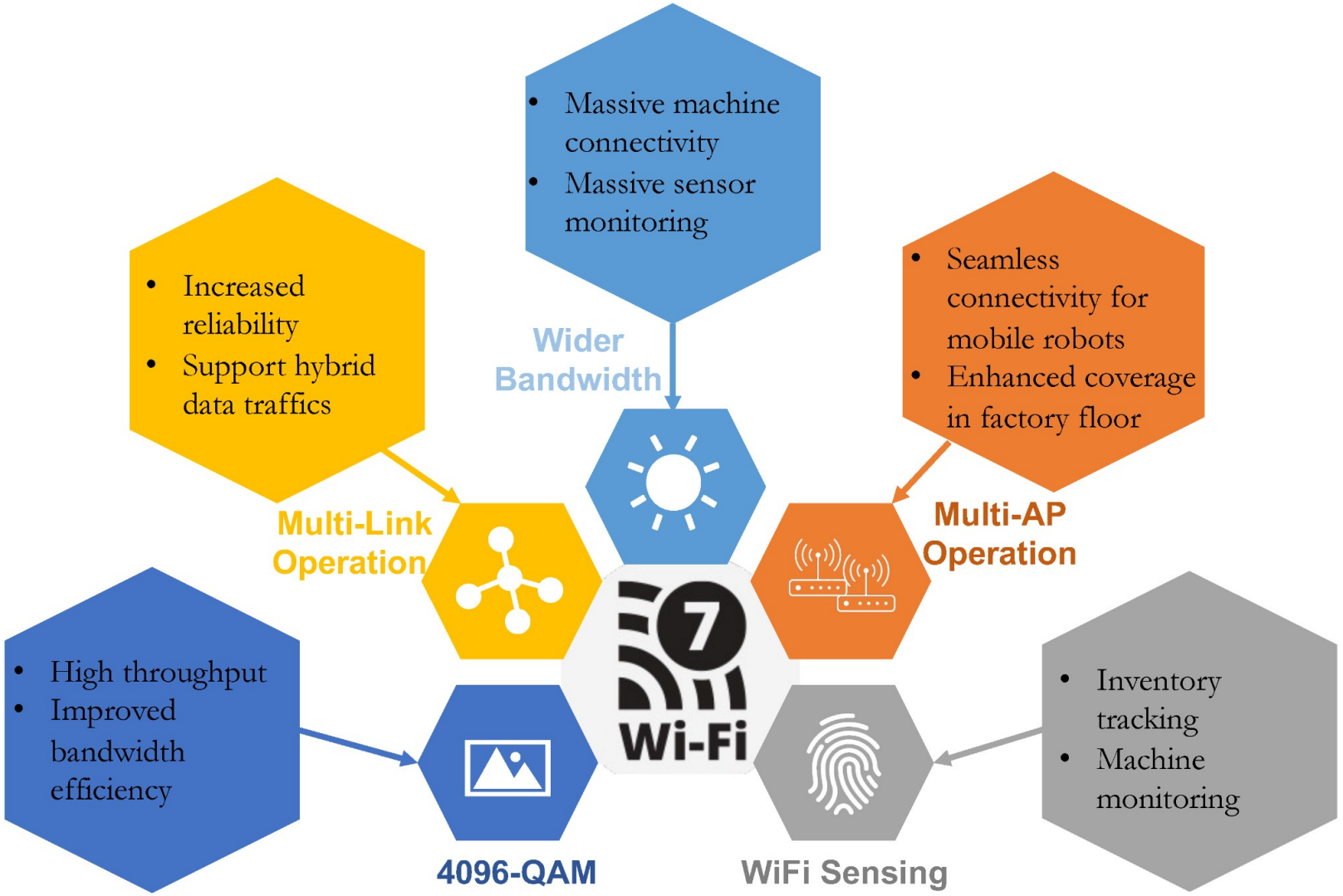
WiFi 7 (IEEE 802.11be) Overview

- Latest WLAN technology supporting Industry 4.0 applications
- Enhanced data rates, reliability, and efficiency for industrial environments
- Operates across 2.4, 5, and 6 GHz frequency bands
- Provides real-time communication, security, and high throughput

WiFi 7 Core Features

- **4096-QAM Modulation:** Ultra-fast, low-latency data transmission
- **Multi-Link Operation:** Concurrent use of multiple frequency bands
- **Wide Bandwidth:** Additional 1.2 GHz bandwidth available
- **Multi-AP Operation:** Unified network coverage across factory floors
- **WiFi Sensing:** Enables localization, asset tracking, and safety monitoring

WiFi 7 Applications



WiFi 7 Limitations

- WiFi 7 signal range reduced due to higher modulation and carrier frequency
- Difficulties in certain industrial environments where machines are spread out over large areas
- Standard hasn’t been ratified yet

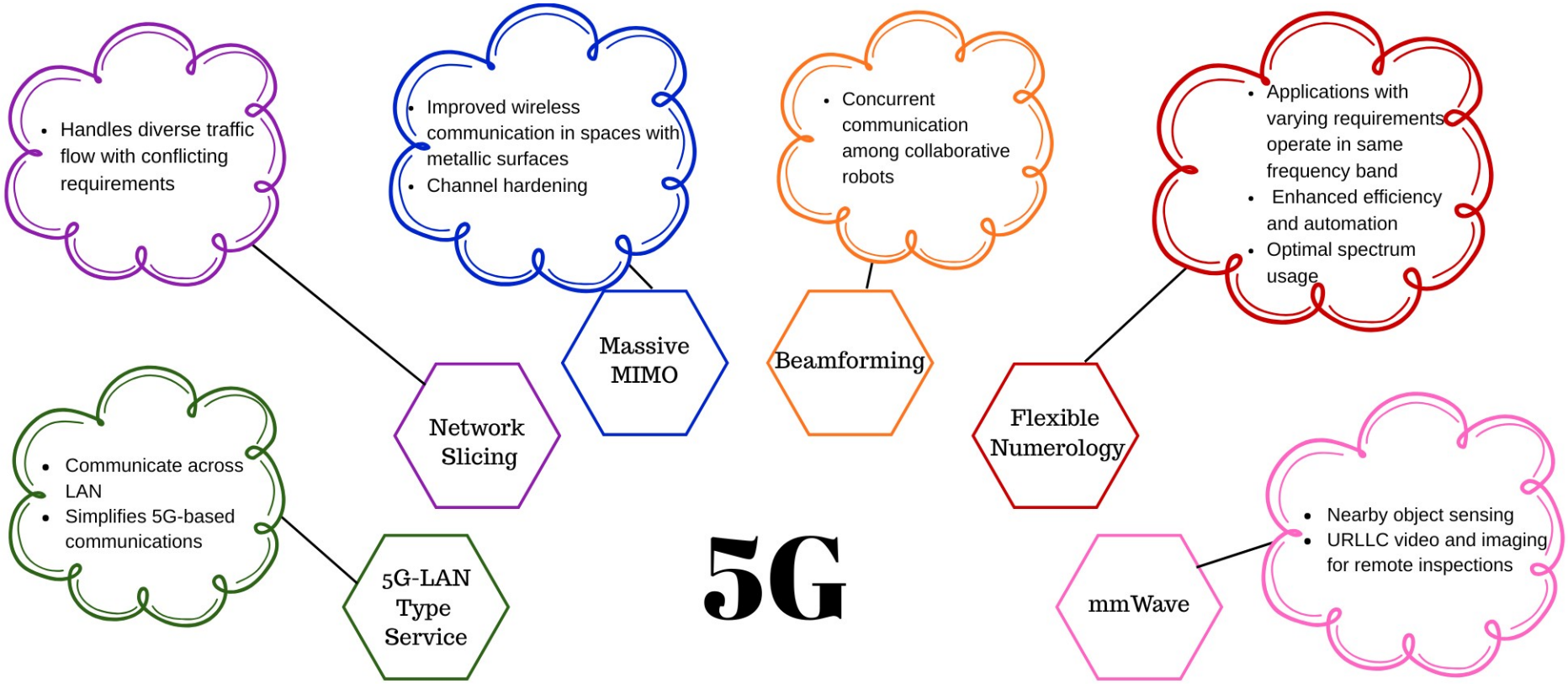
5G Overview

- Varied usage scenarios and applications supported
- Supports four service categories
  - Enhanced mobile broadbands
  - Massive machine-type
  - Fixed wireless access
  - Ultra low-latency communications
- Two main frequency ranges
  - Sub-6 GHz band and mmWave band

5G Core Features

- **Flexible Numerology:** Two main frequency ranges FR1 (sub-6 GHz) and FR2 (mmWave)
- **mmWave:** Provides more bandwidth than sub-6 GHz
- **Beamforming:** Directed beam generated by multiple antenna elements
- **Massive MIMO:** Uses multiple antennas to support multiple device communication simultaneously
- **Network Slicing:** Single physical network divided into multiple virtual networks
- **5G LAN-Type Service:** Replicates LAN features

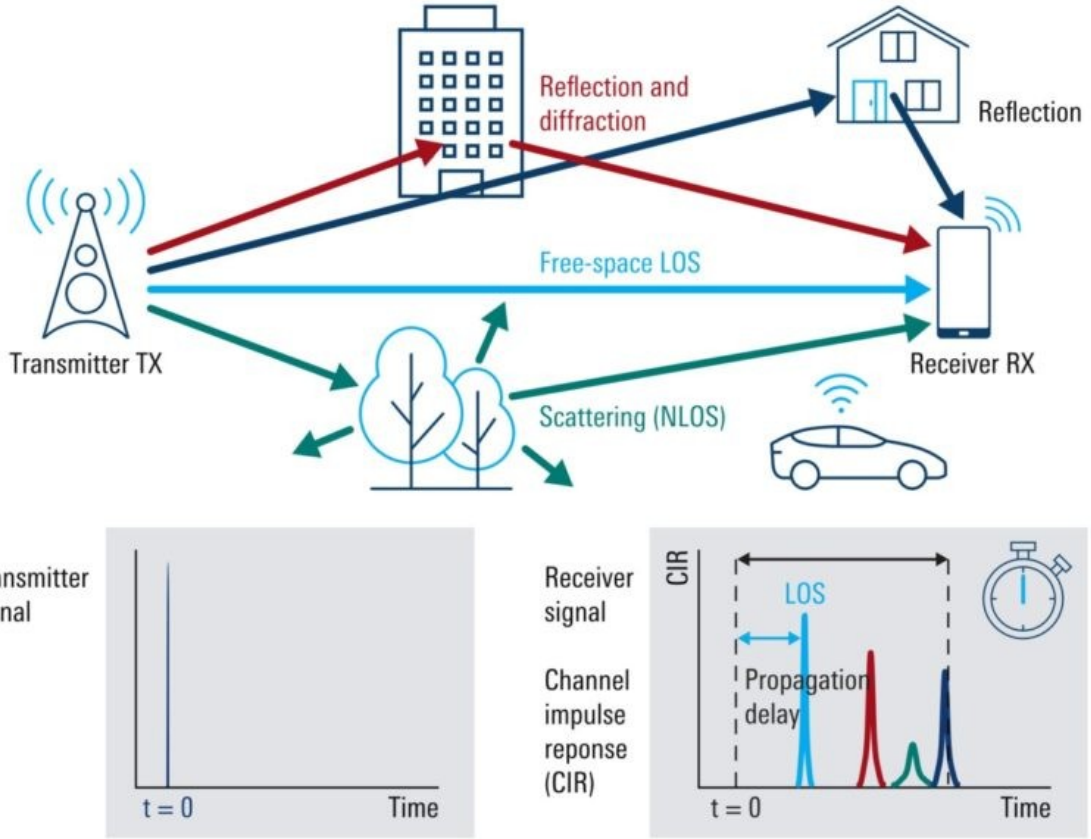
5G Applications



5G Limitations 1/2

Propagation Challenges: (Machinery Interference)

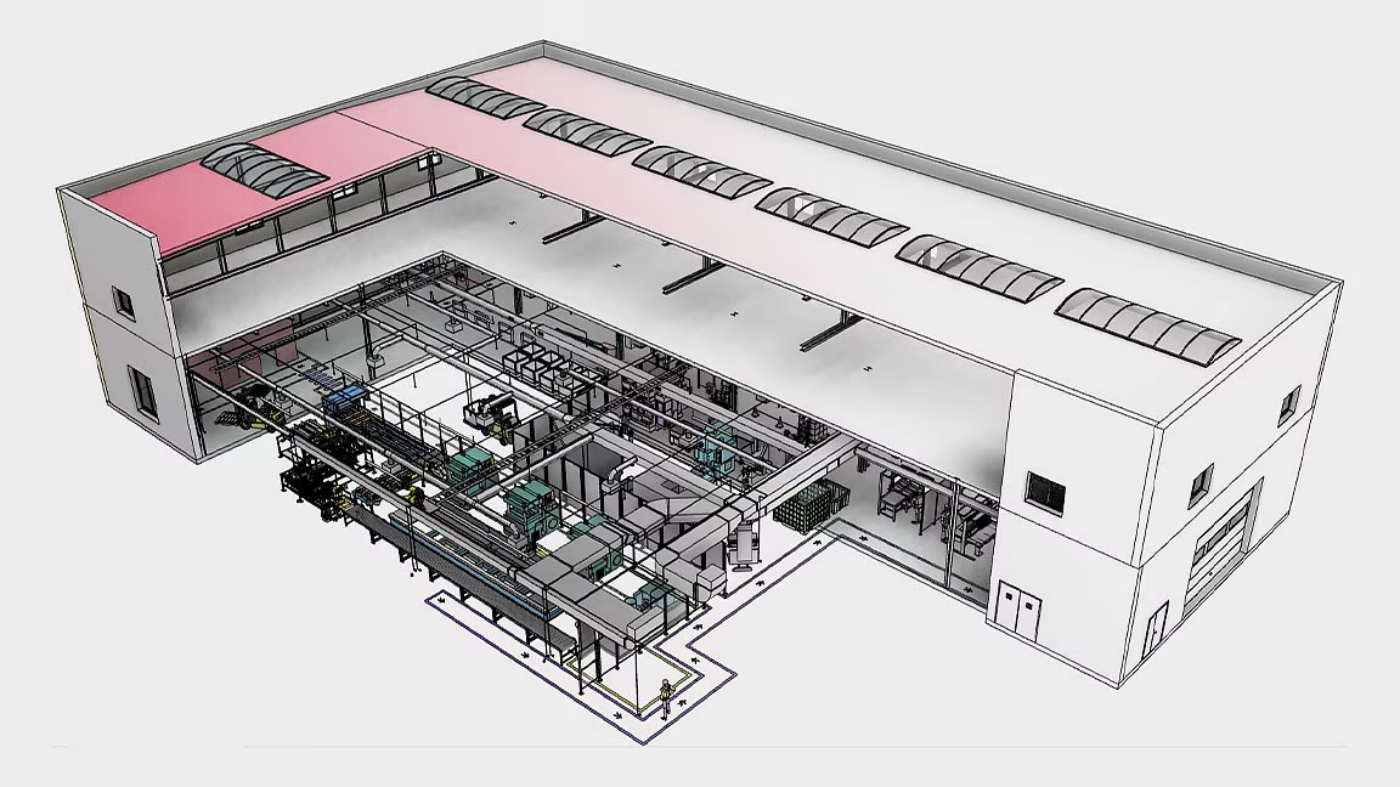
- Shadowing, Scattering, and Reflection



5G Limitations 2/2

Deployment Challenges:

- Private Network Costs
- Planning Issues



WiFi 7 and 5G comparison

Feature	5G	WiFi 7
Implementation Cost	More efficient cost at scale	More cost-effective for localized deployments
Coverage & Mobility	Broad coverage, licensed spectrum, smooth mobility	Shorter range, mostly localized, flexible deployment
Time-Sensitive Networking (TSN)	Best for mission-critical, ultra-low latency applications	Improves reliability in moderate latency/time-sensitive scenarios
Deployment Flexibility	Requires infrastructure planning	Easily deployed/adapted to existing spaces
Hybrid Integration	Supports interoperability in hybrid wired-wireless systems	Works well in hybrid smart factory setups

Challenges and Future Directions 1/2

Augmented Reality (AR)

- Requires ultra high-speed, low-latency networks to deliver real-time information to workers.

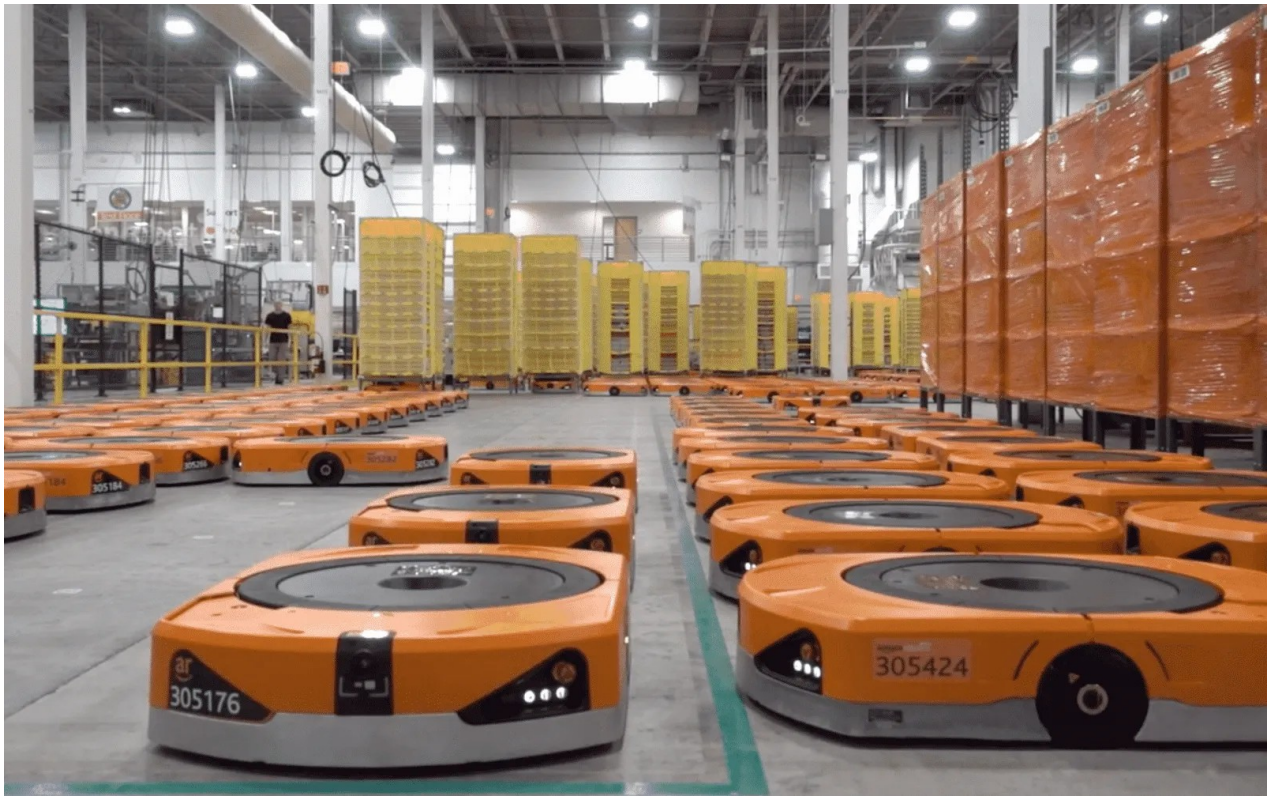
  Any lag can disrupt decision-making or workflow or cause discomfort.



Challenges and Future Directions 2/2

Human-Robot Codependency

- Robots collaborating directly with humans on the factory floor.
  Seamless integration with AI platforms is critical for safety and efficiency.



Deployment Challenges in CPN Networks

5G Deployment

- Optimal placement of base stations is critical to achieve full coverage, low latency, and high reliability.
- Poor planning can lead to blind spots, interference, or degraded performance.

WiFi 7 Spectrum Limitations

- Operates in unlicensed spectrum; may face congestion or interference from existing networks.
- Careful planning needed to ensure predictable performance in dense industrial environments.

Summary

- 5G and WiFi 7 complement each other in smart manufacturing environments
- Integration with TSN ensures deterministic, synchronized operations. Both technologies are key enablers of Industry 4.0 and 5.0 transformation from a connectivity perspective

Without improved communication technologies and reliable networking, future cyber physical systems will face technical limitations.

Future direction: Increasing human-centered aspects of industrial systems (AR/VR), promoting sustainability of systems in CPNs, and including AI connectivity and tooling.

Questions?